COUNTABLE LINEAR ORDERS OF EXACT SIGMA_(LAMBDA+2) SCOTT-SENTENCE COMPLEXITY AT EVERY NONZERO LIMIT LAMBDA

DAVID GONZALEZ, DINO ROSSEGGER

Editorial note. All ordinals are countable; lambda is a nonzero limit and delta_n is a cofinal fundamental sequence.

Editorial note. The summation variable in1403/1435 is read consistently with the explicitly defined delta_n fundamental sequence, and the bounded tail in1437 uses delta_n. Original source notation is preserved.

Editorial note. This construction is not a sum of two shuffle sums. The paper explicitly distinguishes it at1425-1426 and1520-1522 from the already prepared lambda+3 construction; its approximation-sequence and distinguished-cut argument is new work.

Editorial note. The public Z-power back-and-forth lemma141394 is complete support, not a newly counted target.

Editorial note. 2026-10-11 材料收口：复用既有 H2、来源许可和近邻判断；核对原命题与作者证明起止、所指上下文及装配范围齐备。未重证作者结论；沿用既有非重复依据。

Editorial scope. One original exact result. Native construction and load-bearing local dependencies preserved; named classical Scott-analysis inputs explicit.

Allowed standard prerequisites (editorial).

- Standard infinitary Scott analysis. Orbit-definability/Scott-sentence complexity criteria, interval back-and-forth, and the equality of parameterized rank with the maximum ranks of intervals separated by named points.
- Classical scattered-order rank bounds. For a scattered order of bounded Hausdorff rank, the standard Alvir-Rossegger/Ash Scott bounds; a parameter-definable interval cannot exceed the ambient Scott rank. Source-specific Z-power comparison and marker construction are provided in full.

In this article we give a comprehensive Scott analysis of the class of linear orderings using three definitions of rank. Given a structure $\mathcal{A}$ the *(parameterless) Scott rank* $SR(\mathcal{A})$, is the least α, such that the automorphism orbits of all tuples in $\mathcal{A}$ are $\Sigma^{\text{in}}_{\alpha}$ definable; the *parameterized Scott rank* $pSR(\mathcal{A})$, is the least ordinal α such that

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there is a parameter $\bar{p} \in A^{<\omega}$ such that $SR((\mathcal{A}, \bar{p})) = \alpha$; and the *Scott sentence complexity* of $\mathcal{A}$, $SSC(\mathcal{A})$ is the least of the complexities Π^{in}_α, $\Sigma^{\text{in}}_\alpha$, $d\text{-}\Sigma^{\text{in}}_\alpha$ such that $\mathcal{A}$ has a Scott sentence of one of these complexities. While the latter is not an

0.1. **Preliminaries.** We assume that the reader is familiar with the basic techniques of Scott analysis and with linear orderings in general. For background we refer to Montalbán's upcoming book [4] and the standard reference by Ash and Knight [1]. Let us highlight the following lemma which we will use as basic tools in our proofs without reference.

Lemma 0.1 ([1, Lemma 15.8])**.** *Suppose $\mathcal{A}, \mathcal{B} \in \text{LO}$. Then $\mathcal{A} \leq_1 \mathcal{B}$ if and only if $\mathcal{A}$ is infinite or at least as large as $\mathcal{B}$. For $\alpha > 1$, $\mathcal{A} \leq_\alpha \mathcal{B}$ if and only if for every $1 \leq \beta < \alpha$ and every partition of $\mathcal{B}$ into intervals $\mathcal{B}_0, \ldots, \mathcal{B}_n$, with endpoints in $\mathcal{B}$, there is a partition of $\mathcal{A}$ into intervals $\mathcal{A}_1, \ldots, \mathcal{A}_n$ with endpoints in $\mathcal{A}$, such that $\mathcal{B}_i \leq_\beta \mathcal{A}_i$.*

Notice that this lemma particularly implies that $(\mathcal{A}, \bar{a}) \leq_\alpha (\mathcal{B}, \bar{b})$ if and only if $(-\infty, a_0) \leq_\alpha (-\infty, b_0)$, $(a_{|\bar{a}-1|}, \infty) \leq_\alpha (b_{|\bar{b}-1|}, \infty)$ and for all $0 \leq i < |\bar{a} - 1|$ $(a_i, a_{i+1}) \leq_\alpha (b_i, b_{i+1})$ [1, Lemma 15.7].

In the proof of the below lemma we use ζ_α to denote the unique up to isomorphism initial segment of $\mathbb{Z}^\alpha$. It is easy to check that ζ^*_α is the unique end segment of $\mathbb{Z}^\alpha$ and that $\zeta^*_{\alpha+1} = \zeta_\alpha + \mathbb{Z}^\alpha \cdot \omega$.

Theorem 0.2. *For any limit ordinal λ, there is a linear order of Scott sentence complexity $\Sigma^{\text{in}}_{\lambda+2}$.*

Lemma 0.3. *For the sake of organization, consider the following ordinal indexed propositions.*

- (A_α) *For any $\mathcal{K}$ and $\mathcal{L}$, $\mathbb{Z}^\alpha \cdot \mathcal{K} \leq_{2\alpha} \mathbb{Z}^\alpha \cdot \mathcal{L}$.*
- (B_α) *For any $\mathcal{K}$ and $\mathcal{L}$ with $|\mathcal{K}| \geq |L|$, $\mathbb{Z}^\alpha \cdot \mathcal{K} \leq_{2\alpha+1} \mathbb{Z}^\alpha \cdot \mathcal{L}$.*
- (C_α) *For any $\mathcal{K}$ and any $\mathcal{L}$ without a last element, $\mathbb{Z}^\alpha \cdot (\omega + \mathcal{K}) \leq_{2\alpha+2} \mathbb{Z}^\alpha \cdot (\omega + \mathcal{L})$.*

For any countable ordinal α, A_α, B_α and C_α are true.

证明. The proof is by induction with the statements A_0 and B_0 trivially true. We start by showing that B_α implies $A_{\alpha+1}$.

Select a tuple $a_0 \cdots a_n$ in $\mathbb{Z}^{\alpha+1} \cdot \mathcal{L} = \mathbb{Z}^\alpha \cdot \mathbb{Z} \cdot \mathcal{L}$. We will find a tuple $b_0 \ldots b_n$ in $\mathbb{Z}^{\alpha+1} \cdot \mathcal{K} = \mathbb{Z}^\alpha \cdot \mathbb{Z} \cdot \mathcal{K}$ such that $(\mathbb{Z}^{\alpha+1} \cdot \mathcal{L}, \bar{a}) \leq_{2\alpha+1} (\mathbb{Z}^{\alpha+1} \cdot \mathcal{K}, \bar{b})$. Pick a point b_0 arbitrarily. Given b_j, we pick b_{j+1} as follows. If a_j and a_{j+1} are in the same copy of $\mathbb{Z}^{\alpha+1}$, we can pick b_{j+1} such that $(a_j, a_{j+1}) \cong (b_j, b_{j+1})$. If not, we pick b_{j+1} arbitrarily in the copy of $\mathbb{Z}^\alpha$ that is the successor of the successor of the copy of b_j. In other words, we pick b_{j+1} with $(b_j, b_{j+1}) \cong \zeta^*_\alpha + \mathbb{Z}^\alpha + \zeta_\alpha$.

To show that our choice of $\bar{b}$ is correct, we have to confirm the following relations between non-isomorphic intervals.

(1) $(b_j, b_{j+1}) \cong \zeta_\alpha^* + \mathbb{Z}^\alpha + \zeta_\alpha \geq_{2\alpha+1} \zeta_\alpha^* + \mathbb{Z}^\alpha \cdot \mathcal{L}_1 + \zeta_\alpha$, where $\mathcal{L}_1$ is an infinite order,
(2) $(-\infty, b_0) \cong \mathbb{Z}^\alpha \cdot \mathcal{K}_1 + \zeta_\alpha \geq_{2\alpha+1} \mathbb{Z}^\alpha \cdot \mathcal{L}_1 + \zeta_\alpha$, where $\mathcal{L}_1$ and $\mathcal{K}_1$ are some initial segments of the $\mathbb{Z}^\alpha$, and therefore infinite, or
(3) $(b_n, \infty) \cong \zeta_\alpha^* + \mathbb{Z}^\alpha \cdot \mathcal{L}_1 \geq_{2\alpha+1} \zeta_\alpha^* + \mathbb{Z}^\alpha \cdot \mathcal{K}_1$, where $\mathcal{L}_1$ and $\mathcal{K}_1$ are some final segments of the $\mathbb{Z}^\alpha$, and therefore infinite.

It follows immediately from B_α that all of these back-and-forth relations hold.

We now show that B_α implies C_α. So assume that $\mathcal{K}$ and $\mathcal{L}$ have no last element and that B_α holds. Select a tuple $a_0 \cdots a_n$ in $\mathbb{Z}^\alpha \cdot (\omega + \mathcal{L})$. Without loss of generality, a_0 is in the first copy of $\mathbb{Z}^\alpha$. We will find a tuple $b_0 \ldots b_n$ in $\mathbb{Z}^\alpha \cdot (\omega + \mathcal{K})$ such that $(\mathbb{Z}^\alpha \cdot (\omega + \mathcal{L}), \bar{a}) \leq_{2\alpha+1} (\mathbb{Z}^\alpha \cdot (\omega + \mathcal{K}), \bar{b})$. For the a_j in the initial $\mathbb{Z}^\alpha \cdot \omega$, the b_j are picked isomorphically. We now define the rest of the b_j by induction so that all of them are in the initial $\mathbb{Z}^\alpha \cdot \omega$. If a_j is in the same $\mathbb{Z}^\alpha$ block as a_{j+1} or a_{j+1} is in the succesor of a_j's block, define b_{j+1} so that $(b_j, b_{j+1}) \cong (a_j, a_{j+1})$. Otherwise, define b_{j+1} as being some element in the successor of the successor of the $\mathbb{Z}^\alpha$ block of b_j. Not including isomorphic intervals, we only need to check the following cases.

(1) $(b_j, b_{j+1}) \cong \zeta_\alpha^* + \mathbb{Z}^\alpha + \zeta_\alpha \geq_{2\alpha+1} \zeta_\alpha^* + \mathbb{Z}^\alpha \cdot \mathcal{L}_1 + \zeta_\alpha$ where $\mathcal{L}_1$ is infinite,
(2) $(b_n, \infty) \cong \zeta_\alpha^* + \mathbb{Z}^\alpha \cdot (\omega + \mathcal{K}) \geq_{2\alpha+1} \zeta_\alpha^* + \mathbb{Z}^\alpha \cdot \mathcal{L}_1$, where $\mathcal{L}_1$ is some final segment of $\omega + \mathcal{L}$, and thus infinite.

All of these relations follow from B_α, as desired.

To complete the proof of the successor step of the induction, we show that C_α and $A_{\alpha+1}$ imply $B_{\alpha+1}$.

Select a tuple $a_0 \cdots a_n$ in $\mathbb{Z}^{\alpha+1} \cdot \mathcal{L}$. These points belong to at most n distinct copies of $\mathbb{Z}^{\alpha+1}$. Because $|\mathcal{K}| \geq |\mathcal{L}|$, we can match each chosen copy of $\mathbb{Z}^{\alpha+1}$ to one in $\mathbb{Z}^{\alpha+1} \cdot \mathcal{K}$. Furthermore, if there is a copy of $\mathbb{Z}^{\alpha+1}$ between two of the chosen copies in $\mathbb{Z}^{\alpha+1} \cdot \mathcal{L}$, we can assure that the corresponding copies in $\mathbb{Z}^{\alpha+1} \cdot \mathcal{K}$ are also not successors. Also, if a_n is not in the last copy of $\mathbb{Z}^{\alpha+1}$, we can assure that neither is b_n and, similarly for a_0 and b_0 in the first copy. In particular, not including the isomorphic intervals, we only need to check the following cases.

(1) $(b_j, b_{j+1}) \cong \zeta_{\alpha+1}^* + \mathbb{Z}^{\alpha+1} \cdot \mathcal{K}_1 + \zeta_{\alpha+1} \geq_{2\alpha+2} \zeta_{\alpha+1}^* + \mathbb{Z}^{\alpha+1} \cdot \mathcal{L}_1 + \zeta_{\alpha+1}$ for some orders $\mathcal{K}_1$ and $\mathcal{L}_1$,
(2) $(b_n, \infty) \cong \zeta_{\alpha+1}^* + \mathbb{Z}^{\alpha+1} \cdot \mathcal{K}_1 \geq_{2\alpha+2} \zeta_{\alpha+1}^* + \mathbb{Z}^{\alpha+1} \cdot \mathcal{L}_1$ for some orders $\mathcal{K}_1$ and $\mathcal{L}_1$,
(3) $(-\infty, b_0) \cong \mathbb{Z}^{\alpha+1} \cdot \mathcal{K}_1 + \zeta_{\alpha+1} \geq_{2\alpha+2} \mathbb{Z}^{\alpha+1} \cdot \mathcal{L}_1 + \zeta_{\alpha+1}$ for some orders $\mathcal{K}_1$ and $\mathcal{L}_1$,
(4) $(-\infty, b_0) \cong \zeta_{\alpha+1} \geq_{2\alpha+2} \mathbb{Z}^{\alpha+1} \cdot \mathcal{L}_1 + \zeta_{\alpha+1}$ for some order $\mathcal{L}_1$,
(5) $(b_n, \infty) \cong \zeta_{\alpha+1}^* + \mathbb{Z}^{\alpha+1} \cdot \mathcal{K}_1 \geq_{2\alpha+2} \zeta_{\alpha+1}^*$ for some order $\mathcal{K}_1$.

The first three cases are handled immediately by $A_{\alpha+1}$. By symmetry it is sufficient to show case (5). However, this is the same as showing that

$$\zeta_\alpha^* + \mathbb{Z}^\alpha \cdot \omega \leq_{2\alpha+2} \zeta_\alpha^* + \mathbb{Z}^\alpha \cdot (\omega + \mathbb{Z} \cdot \mathcal{K}_1).$$

This follows directly from C_α.

At last we consider the remaining limit levels. We already have seen that B_λ implies C_λ. The statement A_λ follows immediately from A_γ for $\gamma < \lambda$. A bit more nuanced is the issue of B_λ. It follows from A_λ and C_γ for $\gamma < \lambda$. Analogous to the proof in the successor case, we analyze the possibilities:

(1) $\zeta_\lambda^* + \mathbb{Z}^\lambda \cdot \mathcal{K}_1 + \zeta_\lambda \geq_\lambda \zeta_\lambda^* + \mathbb{Z}^\lambda \cdot \mathcal{L}_1 + \zeta_\lambda$ for some orders $\mathcal{L}_1$ and $\mathcal{K}_1$,
(2) $\zeta_\lambda^* + \mathbb{Z}^\lambda \cdot \mathcal{K}_1 \geq_\lambda \zeta_\lambda^* + \mathbb{Z}^\lambda \cdot \mathcal{L}_1$ for some orders $\mathcal{L}_1$ and $\mathcal{K}_1$,
(3) $\mathbb{Z}^\lambda \cdot \mathcal{K}_1 + \zeta_\lambda \geq_\lambda \mathbb{Z}^\lambda \cdot \mathcal{L}_1 + \zeta_\lambda$ for some orders $\mathcal{L}_1$ and $\mathcal{K}_1$,
(4) $\mathbb{Z}^\lambda \cdot \mathcal{K}_1 + \zeta_\lambda \geq_\lambda \zeta_\lambda$ for some order $\mathcal{K}_1$,
(5) $\zeta_\lambda^* + \mathbb{Z}^\lambda \cdot \mathcal{K}_1 \geq_\lambda \zeta_\lambda^*$ for some order $\mathcal{K}_1$.

The first three cases are handled immediately by A_λ. By symmetry it is clear that it is enough to show case 5. However, this is the same as showing that for all $\gamma < \lambda$,

$$\zeta_\gamma^* + \mathbb{Z}^\gamma \cdot (\omega + \mathbb{Z} \cdot \zeta_\lambda^*) \leq_\gamma \zeta_\gamma^* + \mathbb{Z}^\gamma \cdot (\mathbb{Z} \cdot \zeta_\lambda^* + \mathbb{Z}^\lambda \cdot \mathcal{K}_1),$$

which follows from the C_γ for $\gamma < \lambda$.

Therefore, for any countable ordinal α, A_α, B_α and C_α all hold. □

Proposition 0.4. *For any limit ordinal λ, there is a scattered linear order of Scott sentence complexity Π^{in}_λ.*

证明. Let $(\delta_n)_{n\in\omega}$ be a fundamental sequence for λ. We show that

$$\mathcal{L}_\lambda := \sum_{i\in\omega} i + \mathbb{Z}^{\delta_n}$$

is of the desired complexity.

First note that it indeed does have a Π^{in}_λ Scott sentence. The sentence states the following:

(1) There is exactly one 1-block isomorphic to each natural number.
(2) These 1-blocks are ordered like the natural numbers.
(3) The order between n and $n+1$ is isomorphic to $\mathbb{Z}^{\delta^n}$.

This is Π^{in}_λ because each of the items is $\Pi^{\text{in}}_{<\lambda}$. The first two statements can be expressed using finitely many alternations of quantifiers. The final statement can be expressed by the Scott sentence of $\mathbb{Z}^{\delta_n}$ relativized to the specific interval, this sentence has Scott sentence complexity below λ by [2], as the Hausdorff rank of the interval is below λ.

By Lemma 0.3, the $\mathbb{Z}^{\delta_n}$ have unbounded Scott rank below λ. As these are all Δ^{in}_0-definable over parameters in $\mathcal{L}_\lambda$, by [6, Lemma 4.3], we have that for each n, $SR(\mathcal{L}_\lambda) \geq \delta_n$, so Π^{in}_λ is indeed the optimal Scott sentence complexity. □

证明. Let $(\delta_n)_{n\in\omega}$ be a fundamental sequence for λ and let

$$\mathcal{L}_\lambda := \sum_{i\in\omega} i + \mathbb{Z}^{\delta_n}.$$

Furthermore, define for every n, the linear orderings:

$$\mathcal{L}_{\lambda,n} := \sum_{i<n} i + \mathbb{Z}^{\delta_i} + \sum_{i\geq n} i + \mathbb{Z}^{\delta^n}.$$

Note that by Lemma 0.3, $\mathcal{L}_\lambda$ is approximated by the sequence of these orderings in the sense that $\mathcal{L}_\lambda \equiv_{2\delta_n} \mathcal{L}_{\lambda,n}$.

Let $\mathcal{B} = \mathcal{L}_\lambda \cdot (1+\mathbb{Q})$ and $\mathcal{A}$ be a copy of $\mathbb{Q}$ where an unbounded, increasing sequence of points c_n is replaced by $\mathcal{L}_{\lambda,n}$ and each other point is replaced by $\mathcal{L}_\lambda$. We now claim that $\mathcal{K} = \mathcal{A} + \mathcal{B}$ is the desired order. Note that over the parameter of the initial element of $\mathcal{B}$, $SR_p(\mathcal{K}) = \max(SR(\mathcal{A}), SR(\mathcal{B}))$ (this equality follows from [5, Lemma 11]). We now demonstrate that this quantity is exactly λ.

To define the automorphism orbits of elements within $\mathcal{B}$ we have to distinguish elements in the initial copy of $\mathcal{L}_\lambda$. For an element $b \in \mathcal{B}$, let n be the size of the finite block to the left of b. We can write a formula of finite rank saying that there is precisely one finite block of size n to the left of b and thus use this type of formulas to distinguish elements in the first $\mathcal{L}_\lambda$ copy. If b is in a finite block we conjunct this formula or its negation with its position in this finite block, if b is in a copy of $\mathbb{Z}^{\delta_n}$, then we relativize the formula defining its automorphism orbit in this ordering to the interval restricted by block of size n to the left and a block of size $n+1$ to the right, just as in the proof of Proposition 0.4. Conjuncting this with the formula deciding whether b is in the initial copy of $\mathcal{L}_\lambda$ yields the defining formula for its automorphism orbit. To extend this definition to tuples we take the conjunction of the formulas defining orbits for elements and a formula specifying the order of the tuple.

For $\mathcal{A}$ the definitions are a little more subtle, but generally follow the same plan. In particular, if the point in $\mathbb{Q}$ that x lies in is between c_n and c_{n+1}, then the following describes the automorphism orbit of x:

(1) Let z be the greatest element below x with no successor or predecessor.
(2) Let z_n be the initial element with no successor or predecessor within c_n; i.e. say that it is the greatest element with no successor or predecessor below an $n+1$ and $n+2$ block that bound a structure isomorphic to $\mathbb{Z}^{\delta_n}$.
(3) Define z_{n+1} analogously to z_n, expect it is in c_{n+1}.
(4) Say that $z_n \leq x < z_{n+1}$.
(5) Check if $z = z_n$ or not.
(6) In an analogous manner to the case of $\mathcal{A}$, define the automorphism orbit of x within its copy of $\mathbb{Z}^{\delta_k}$ or the finite block that it is in.

This describes the automorphism orbit of a point and is of complexity less than $\Sigma^{\text{in}}_\lambda$. In order to extend this definition to tuples the only additional detail needed

is the order of the tuple and whether the tuples lie in the same point in $\mathbb{Q}$ or not. This can be done by comparing the various values of "z" that emerge for each point in the tuple.

In order to finish the proof, all that remains is to show that $SR(\mathcal{K}) = \lambda + 2$, or in other words, that it has an orbit that it is not $\Sigma^{\text{in}}_{\lambda+1}$-definable. Let c be the initial point in $\mathcal{B}$ and b be some other point in $\mathcal{B}$ in a block of size 1. We will show that $(\mathcal{K}, b) \leq_{\lambda+1} (\mathcal{K}, c)$, so the orbit of c is not $\Sigma^{\text{in}}_{\lambda+1}$ definable in $\mathcal{K}$.

Note that $\mathcal{K}_{>c} \cong \mathcal{K}_{>b}$, so it is sufficient to show that $\mathcal{K}_{<b} \leq_{\lambda+1} \mathcal{K}_{<c}$. As $\mathcal{K}_{<c}$ is an initial segment of $\mathcal{K}_{<b}$, for the first turn of the game, the $\exists$-player can pick whatever points the $\forall$-player picked in $\mathcal{K}_{<c}$ according to the canonical embedding of $\mathcal{K}_{<c}$ in $\mathcal{K}_{<b}$. Note that each interval is isomorphic except for the one between the greatest chosen points (call them a_k) and c or b respectively. a_k may be greater than some finite number of the c_n. Therefore, up to a finite difference in the choice of sequence δ_n used to define $\mathcal{L}_{\lambda,n}$, (a_k, c) is just $\mathcal{M} + \mathcal{A}$ where $\mathcal{M}$ is $\mathcal{L}_{\lambda,>c}$ or $\mathcal{L}_{\lambda,n,>c}$ depending on which type of order a_k lives in and (a_k, b) is just $\mathcal{M} + \mathcal{K}$. In short, it is enough to demonstrate that

$$\mathcal{A} \geq_\lambda \mathcal{K},$$

to show the claim. To see this, choose $\alpha < \lambda$ and show that $\mathcal{A} \geq_\alpha \mathcal{K}$. Take $\delta_n > \alpha$. It is indeed the case that for $k \geq n$, $\mathcal{L}_\lambda \equiv_\alpha \mathcal{L}_{\lambda,k}$. For some point d in a block of size 1 between c_n and c_{n+1} we have that

$$\mathcal{A} = \mathcal{A}_{<d} + \sum_{q\in\mathbb{Q}} \mathcal{A}_q \quad \text{and} \quad \mathcal{K} = \mathcal{A}_{<d} + \sum_{q\in\mathbb{Q}} \mathcal{B}_q,$$

where $\mathcal{A}_q \equiv_\alpha \mathcal{L}_\lambda \equiv_\alpha \mathcal{B}_q$. This proves the claim. □

References

[1] C. Ash and J. Knight. *Computable Structures and the Hyperarithmetical Hierarchy*. Studies in Logic and the Foundations of Mathematics144,2000.

[2] R. Alvir and D. Rossegger. The complexity of Scott sentences of scattered linear orders. *Journal of Symbolic Logic*85(3)(2020),1079–1101.

[3] C.J. Ash. Recursive labelling systems and stability of recursive structures in hyperarithmetical degrees. *Transactions of the American Mathematical Society*298(2)(1986),497–514.

[4] A. Montalban. *Computable Structure Theory: Beyond the Arithmetic*. Draft,2021. `https://math.berkeley.edu/~antonio/CSTpart2_DRAFT.pdf`.

[5] David Gonzalez and Antonio Montalb'an. *The ω-Vaught's Conjecture*. Native bibliography entry supplied in flattened.bbl.

[6] Antonio Montalb'an. Classes of structures with no intermediate isomorphism problems. *Journal of Symbolic Logic*81 (2016),127–150.